



Pradeep Kallathuru

Azure Snowflake Data Engineer

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PROFESSIONAL SUMMARY:

- Certified Senior Data Engineer with 12+ years of experience in Data Engineering with strong foundation in relational database concepts, data modeling, and cloud-based data warehousing. I have 5 years of expertise in Data Warehousing and ETL processes, working extensively with Azure cloud services, Big Data technologies, and Snowflake.
- With Proficient in designing and implementing ETL/ELT workflows to extract, transform, and load data into Snowflake, data lakes, and cloud data warehouses on Azure and AWS.
- Hands-on expertise in Spark-based processing for large-scale data transformations, performance tuning, and batch/real-time processing using Spark Streaming and Kafka.
- Strong command of data modeling, schema design, and optimization techniques such as partitioning, clustering, and caching to improve pipeline efficiency.
- Experienced in implementing data quality validation checks and monitoring processes to ensure reliable and accurate data delivery.
- Proven ability to collaborate with cross-functional teams, including analysts, data scientists, and DevOps engineers, to translate business requirements into scalable technical solutions.
- Deep experience working in cloud environments (Azure and AWS), with practical knowledge of CI/CD automation, data governance, and cost optimization.
- Adaptable and proactive professional who keeps pace with emerging big data technologies and industry trends.
- Experience in implementing OLAP multi-dimensional cube functionality using Azure SQL Data Warehouse, Azure Data Factory, ADLS, Databricks, and SQL DW.
- Strong experience using Spark RDD API, Spark Data frame/Dataset API, Spark-SQL, and Spark ML frameworks for building end-to-end data pipelines.
- Experience with Python, and SQL on Azure cloud platform, a better understanding of Data Warehouses like Azure Synapse and Azure Data-bricks platform, etc.
- Strong expertise in optimizing Spark jobs and leveraging Azure Synapse Analytics for big data processing and analytics.
- Experienced in Azure Data Factory with Azure Logic Apps for orchestrating complex data workflows and triggering actions.
- Experienced in developing CI/CD frameworks for data pipelines and collaborating with DevOps teams for automated pipeline deployment.
- Good experience working with real-time streaming pipelines using Kafka and Spark-Streaming.
- Proficient in leveraging Azure Databricks and Spark for distributed data processing and transformation tasks.
- Skilled in ensuring data quality and integrity through validation, cleansing, and transformation operations.
- Adept at designing cloud-based data warehouse solutions using Snowflake on Azure, optimizing schemas, tables, and views for efficient data storage and retrieval.
- Experience with Snowflake Multi-Cluster Warehouses.
- Understanding of Snowflake cloud technology.
- Experience in using Snowflake Clone and Time Travel.
- Participates in the development, improvement, and maintenance of Snowflake database applications.
- Build the Logical and Physical data model for Snowflake as per the changes required.
- In-depth knowledge of Snowflake Database, Schema and Table structures.
- Proven track record in performance optimization and capacity planning to ensure scalability and efficiency.
- Proficient in scripting languages such as Python and Scala.
- Skilled in working with Hive, SparkSQL, Kafka, and Spark Streaming for ETL tasks and real-time data processing.
- Have a good working experience in Hadoop, HDFS, Map-Reduce, Hive, Tez, Python, PySpark.
- Hands-on working experience and developing large-scale data pipelines using Spark and Hive.

- Experience in using Apache Sqoop to import and export data from HDFS and Hive.
- Hands-on experience in setting up workflow using Apache Oozie workflow engine for managing Hadoop jobs.
- Experienced in setting up workflow using Apache Oozie workflow engine for managing and scheduling Hadoop jobs and importing and exporting the data using Sqoop from HDFS to Relational Database systems.
- Optimized query performance in Hive using bucketing and partitioning techniques and have extensive hands-on experience tuning Spark Jobs. Implemented data visualization using Power BI & Power BI DAX.
- Hands-on experience in implementing shell scripting.
- Implemented Datawarehouse solutions using Snowflake & Start Schema.
- Implemented data pipelines using Informatica & PL SQL
- Highly proficient in Agile methodologies, including JIRA for project management and reporting.

EDUCATION:

- Bachelor's in electrical and Electronic Engineering from Sreenidhi Institute of Science and Technology Jawaharlal Nehru Technological University, India.

TECHNICAL SKILLS:

GCP, Azure, AWS Cloud Services	Google Cloud Platform (GCP), Google Cloud SQL, BigQuery, Cloud Storage, Cloud Pub/Sub, Kubernetes Engine, App Engine, Compute Engine, Cloud Functions, Cloud Spanner, Cloud Datastore, Cloud Composer, Dataflow, Dataproc, Cloud Monitoring, Cloud Logging, Cloud IAM, Cloud Build, Cloud Source Repositories, Cloud Identity, Cloud Security Command Center, Amazon Health Lake, AWS Glue, Amazon CloudWatch, Amazon Kinesis, Route 53, IAM, EC2, AWS Lambda, Amazon Quick Sight, Amazon Macie, MSK; Azure Data Factory, Azure Data Bricks, snowflake, Logic Apps, Functional Apps, Snowflake, Azure DevOps, Azure Data Lakes, Azure Blob Storage, Azure SQL Database, Azure Synapse Analytics.
Big Data Technologies	HDFS, Yarn, Map Reduce, Hive, SQOOP, Flume, HBase, PySpark, Kafka
Web Technologies	HTML5, CSS3, XML, JDBC, JSP, RestAPI.
Databases	My SQL Server, Teradata, Oracle 11g/12c, MySQL, NoSQL, Cassandra, Cosmos DB, DB2.
Languages	Python, Java, Scala, SQL, PySpark, PL/SQL, HiveQL, T-SQL
Version Control Tools	SVM, GitHub, Bitbucket, GitLab
Hadoop Distribution	Cloudera, HortonWorks
Visualization Tools	Power BI, Tableau, MicroStrategy
ETL Tools	Informatica, SSIS, SSRS.
Operating Systems	Windows (XP/7/8/10), UNIX, LINUX, UBUNTU, CENTOS.
Build Automation Tools	Ant, Maven
IDE & Build tools	PyCharm, Eclipse, Visual Studio.

Client: CVS- Scottsdale Arizona

Oct 2022 - Till now

Role: Azure Snowflake Data engineer

Responsibilities:

- Designed and implemented scalable ETL/ELT pipelines using Azure Data Factory, Databricks (PySpark, Spark SQL), and Snowflake, efficiently processing large datasets from multiple sources.
- Built robust data ingestion workflows integrating Kafka, ADLS, and Delta Lake, supporting both batch and real-time processing requirements.
- Applied advanced data modeling techniques in Snowflake, leveraging features like clustering keys, partitioning, and caching to improve query performance and storage efficiency.
- Developed automated data quality checks and monitoring processes using PySpark, ensuring consistency and integrity across pipeline stages.
- Worked closely with data analysts and data scientists to define technical specs from business requirements and enabled self-serve analytics via Power BI integration.
- Optimized Spark jobs using performance tuning techniques such as broadcasting, caching, and predicate pushdown to reduce execution time by over 40%.
- Actively Designed and developed automated data pipelines in Python for large-scale data ingestion and transformation.

- Implemented end-to-end data workflows using advanced shell scripting in UNIX environments.
- Optimized Snowflake data models and queries for faster performance and lower compute costs.
- Managed S3 buckets and utilized AWS services (like Lambda, Glue, EC2) for efficient data storage and processing.
- Collaborated with cross-functional teams using Agile methodologies, with task tracking via JIRA.
- Implemented partitioning, indexing, and caching strategies in Snowflake and Azure services to enhance query performance and reduce processing time.
- Conducted performance tuning and capacity planning exercises to ensure the scalability and efficiency of the data infrastructure.
- Developed CI/CD framework for data pipelines using Jenkins tool.
- Collaborated with DevOps engineers to develop automated CI/CD and test-driven development pipelines using Azure as per the client's requirement.
- Involved in running all the Hive scripts through Hive on Spark and some through Spark SQL.
- Developed a data pipeline using Kafka, Spark, and Hive to ingest, transform and analyze data.
- Designed and implemented real-time data processing solutions using Kafka and Spark Streaming, enabling the ingestion, transformation, and analysis of high-volume streaming data.
- Developed Spark core and Spark SQL scripts using Scala for faster data processing.

Environment: Azure Databricks, Data Factory, Snowflake, Logic Apps, Functional App, Snowflake, MS SQL, Oracle, HDFS, MapReduce, YARN, Spark, Hive, SQL, Python, Scala, PySpark, shell scripting, GIT, JIRA, Jenkins, Kafka, ADF Pipeline, Power BI.

Client: Bank Of America Bank, Scottsdale Arizona

April 2020 – Sep 2022

Role: Azure Snowflake data engineer

Responsibilities:

- Built Engineered and maintained end-to-end big data pipelines across Azure and AWS platforms using ADF, PySpark, Glue, and Redshift, ensuring high-throughput and fault-tolerant data flows.
- Designed a centralized data lake architecture using ADLS Gen2, integrating multiple enterprise data sources with automated ingestion and schema evolution support.
- Implemented ETL workflows in PySpark and SQL to extract, cleanse, transform, and load data into Snowflake, enabling secure and fast access for downstream analytics.
- Collaborated with data science teams to provide well-structured and pre-processed datasets for modeling and advanced analytics use cases.
- Developed real-time streaming pipelines using Kafka and Spark Streaming to support fraud detection use cases and deliver insights within seconds.
- Managed orchestration and job scheduling through Apache Airflow, with alerts and retries built in to minimize downtime and data gaps.
- Conducted regular data profiling, validation, and auditing, contributing to enterprise data governance and compliance.
- Improved performance of long-running Spark jobs by applying tuning strategies such as file format conversion (CSV Parquet), optimized joins, and cluster sizing.
- Integrated Azure Data Factory with Azure Logic Apps for orchestrating complex data workflows and triggering actions based on specific events.
- Developed data ingestion pipelines using Azure Event Hubs and Azure Functions to enable real-time data streaming into Snowflake.
- Designed and implemented data processing workflows using Azure Databricks, leveraging Spark for large-scale data transformations.
- Implemented data governance practices and data quality checks using AWS Glue and Amazon Redshift, ensuring data accuracy and consistency.
- Implemented data replication and synchronization strategies between Snowflake, Amazon Redshift, and other data platforms using AWS Glue and Change Data Capture techniques.
- Developed and deployed AWS Lambda functions and Azure Functions for data preprocessing, data enrichment, and data validation tasks in data pipelines.
- Implemented advanced analytics and machine learning workflows using Amazon Sage Maker, Amazon Redshift, Azure Machine Learning, and Snowflake, enabling predictive analytics and data-driven insights.

- Designed and implemented data archiving and data retention strategies using Azure Blob Storage and Snowflake's Time Travel feature.
- Developed custom monitoring and alerting solutions using Azure Monitor and Snowflake Query Performance Monitoring (QPM) for proactive identification and resolution of performance issues.
- Integrated Snowflake with Power BI and Azure Analysis Services for creating interactive dashboards and reports, enabling self-service analytics for business users.
- Optimized data pipelines and Spark jobs in Azure Databricks for improved performance, including tuning Spark configurations, caching, and leveraging data partitioning techniques.
- Integrated Amazon Redshift with Amazon QuickSight and AWS Glue DataBrew for creating interactive dashboards and reports, enabling self-service analytics for business users.
- Collaborated with cross-functional teams including data scientists, data analysts, and business stakeholders to understand data requirements and deliver scalable and reliable data solutions.

Environment: AWS Glue, Amazon Redshift, AWS EMR, S3, SageMaker, Amazon QuickSight, Azure Databricks, Data Factory, Logic Apps, Snowflake, Functional App, Snowflake, MS SQL, Oracle, HDFS, MapReduce, YARN, Spark, Hive, SQL, Python, Scala, PySpark, shell scripting, GIT, JIRA, Jenkins, Kafka, ADF Pipeline, Power BI.

Client: Ingram Micro, Irvine, CA

Dec 2017 to March 2020

Role: Big Data Developer

Responsibilities:

- Developed and maintained data pipelines using Sqoop, Flume, and Kafka to ingest, transform, and process customer behavioral data for analysis.
- Performed data aggregation and analysis on large-scale datasets using Apache Spark, Scala, and Hive, resulting in improved insights for the business.
- Utilized big data ecosystems such as Hadoop, Spark, and Cloudera to load and transform large sets of structured, semi-structured, and unstructured data.
- Integrated HBase with Hive on the Analytics Zone, creating and optimizing HBase tables for faster and more efficient querying of data.
- Utilized Hive queries and Spark SQL to analyze and process data, meeting specific business requirements and simulating MapReduce functionalities.
- Implemented automation for deployments using YAML scripts, resulting in faster and more efficient builds and releases.
- Migrated data from RDBMS (Oracle) to Hadoop using Sqoop for processing, enhancing data management and processing capabilities.
- Utilized JIRA to manage issues and project workflow, improving project organization and efficiency.
- Collaborated with team members to identify and resolve JVM-related issues, resulting in better performance and system stability.
- Utilized Git as a version control tool to maintain the code repository, ensuring better code management and tracking of changes.

Environment: Sqoop, MySQL, HDFS, Apache Spark Scala, Hive Hadoop, Cloudera, HBase, Kafka, MapReduce, Zookeeper, Oozie, Data Pipelines, RDBMS, AWS, EC2, Python, PySpark, shell script, Ambari, JIRA.

Client: Zillow Group - Seattle, WA

Aug 2016 - Nov 2017

Role: Big Data Developer

Responsibilities:

- Prepared an ETL framework using Sqoop, Pig, and Hive to bring in data from various sources and make it available for consumption.
- Processed HDFS data and created external tables using Hive, along with developing scripts for table ingestion and repair for reuse across the project.
- Developed ETL jobs using Spark and Scala to migrate data from Oracle to new MySQL tables.
- Utilized Spark (RDDs, DataFrames, Spark SQL) and Spark-Cassandra Connector APIs for various tasks, including data migration and business report generation.
- Developed a Spark Streaming application for real-time sales analytics.
- Analyzed source data, efficiently handled data type modifications, and used Excel sheets, flat files, and CSV files to generate Power BI ad-hoc reports.
- Analyzed SQL scripts and designed solutions using PySpark.
- Extracted data from various data sources into HDFS using Sqoop.

- Handled data import from various sources, performed transformations using Hive and MapReduce, and loaded data into HDFS.
- Extracted data from MySQL into HDFS using Sqoop.
- Implemented automation for deployments using YAML scripts for streamlined builds and releases.
- Worked with Apache Hive, Apache Pig, HBase, Apache Spark, Zookeeper, Flume, Kafka, and Sqoop.
- Implemented data classification algorithms using MapReduce design patterns.
- Extensively worked on creating combiners, partitioning, and distributed cache to enhance the performance of MapReduce jobs.
- Utilized Git and GitHub repositories to maintain the source code and enable version control.

Environment: Hadoop, Hive, spark, PySpark, Sqoop, Spark SQL, Shell script, Cassandra, YAML, ETL.

Client: Aetna Inc., Hartford, CT

June 2015 – Aug 2016

Role: Data Warehouse Developer

Responsibilities:

- Experienced in designing ETL flows with SSIS for data extraction from SQL Server and transforming Excel data using MS SQL Server.
- Proficient in extensively writing PL/SQL and SQL programs, and developing views, materialized views, functions, procedures, packages, triggers, and cursors for optimized query performance.
- Expertise in effectively monitoring and optimizing SQL Server Performance, employing industry best practices and techniques to ensure efficient and high-performing database systems.
- Experienced in creating ETL workflows with Informatica and DataStage, ensuring smooth data integration into the warehouse with complex transformations to meet business needs.
- Expertise in finding Facts and Dimensions, creating fact and dimension tables, and managing Slowly Changing Dimensions (SCD) for efficient dimensional data modeling for Data Mart design.
- Experienced in utilizing NoSQL databases MongoDB and Cassandra for handling and analyzing large volumes of unstructured and semi-structured data in data warehouse environments, enabling scalable storage and retrieval.
- Skilled in building Cubes and Dimensions, integrating designs and data sources, and leveraging MDX scripting for advanced analytics.
- Proficient in data mart schema designs like Star Schemas T-SQL schemas and experienced in Python for data tasks and Erwin for warehouse development, ensuring integrity and scalability.
- Proficient in SSRS report development on SSAS Cubes with various functionalities, and skilled in SSAS cube development encompassing aggregation, KPIs, measures, partitioning, and deployment.
- Proficient in data visualization and analysis using Tableau, creating interactive dashboards and reports to effectively communicate insights and support data-driven decision-making.
- Skilled in IBM DB2 ETL development, ensuring smooth data transfer to the warehouse while optimizing workflows for better performance and scalability.
- Extensively worked on Performance Tuning and Query Optimization using distinct types of Hints, Partitioning, Bulking Techniques, and indexes.
- Proficient in SharePoint administration, optimizing solutions for document management, collaboration, and process automation to boost productivity.
- Demonstrated extensive experience in utilizing Unix Shell Scripts for efficient support and server-based tasks.
- Proficient in utilizing GIT as a version control tool for effective code repository maintenance, facilitating superior code management.

Environment: Windows server, MS SQL Server 2014, SSIS, SSAS, SSRS, SQL Profiler, Power BI, C#, Performance Point Server, MS Office, SharePoint

Client: Credit Suisse - New York, NY

Feb 2014 – May 2015

Role: Data Warehouse Developer

Responsibilities:

- Involved in Analysis, Planning and Defining data based on business requirements and provided documentation.
- Created more stored procedures using Common Table Expression (CTE) and various types of UDF.
- functions.
- Used DDL and DML for writing triggers, stored procedures to check the data entry and payment verification.
- Analyzed the existing application programs and tuned SQL queries using Execution plan, SQL Performance Monitor, SQL Profiler.

- Created DDL scripts to create database schema and database objects.
- Expertise in solving any issues related to Microsoft SQL server.
- Build and maintain complex SQL queries using T SQL for data analysis, data mining and data manipulation.
- Organize analytical data ensure data quality, and aggregate data for strategic reporting and performance optimization on database solutions.
- Different database triggers were created and stored in the database and fired off when contents of database were changed.
- Responsible for creating Databases, Tables, Cluster/Non-Cluster Index, Unique/Check Constraints Views, Stored Procedures, Triggers, Rules
- Extensively used joins and sub-queries for complex queries involving multiple tables from different databases.
- Created and reallocated database objects on appropriate file groups for easy maintenance and improved data access performance.
- Developed Scenarios for unit, integration testing to ensure that all the components work correctly when integrated.

Environment: MS SQL Server 2016, Visual Studio 2017/2019, SSIS, SharePoint, MS Access, Team Foundation Server, Git.